

|                                              |                                                       |
|----------------------------------------------|-------------------------------------------------------|
| Program : <b>Diploma in Renewable Energy</b> |                                                       |
| Course Code : <b>3297</b>                    | Course Title: <b>Analog &amp; Optoelectronics Lab</b> |
| Semester : <b>3</b>                          | Credits: 1.5                                          |
| Course Category: <b>Programme Core</b>       |                                                       |
| Periods per week: <b>3 (L: 0 T: 0 P: 3)</b>  | Periods per semester: <b>45</b>                       |

### Course Objectives:

- To develop rectifier circuits using diodes
- To construct amplifier and oscillator circuits using transistors
- To familiarize with various applications of op-amps
- To identify optoelectronic devices and circuits

### Course Prerequisites:

| Topic                                     | Course Code | Course Name                                              | Semester |
|-------------------------------------------|-------------|----------------------------------------------------------|----------|
| Basic knowledge of electronics components |             | Fundamentals of Electrical & Electronics Engineering Lab | 2        |

### Course Outcomes:

On completion of the course student will be able to:

| CO <sub>n</sub> | Description                                           | Duration (Hours) | Cognitive Level |
|-----------------|-------------------------------------------------------|------------------|-----------------|
| CO1             | Build different diode rectifier circuits              | 9                | Applying        |
| CO2             | Construct different amplifier and oscillator circuits | 12               | Applying        |
| CO3             | Develop various op-amp circuits                       | 9                | Applying        |
| CO4             | Identify various Optoelectronic devices               | 9                | Applying        |
|                 | Lab Test                                              | 6                |                 |

## CO – PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             |     |     |     | 3   |     |     |     |
| CO2             |     |     |     | 3   |     |     |     |
| CO3             |     |     |     | 3   |     |     |     |
| CO4             |     |     |     | 3   |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

## Course Outline

| Module Outcomes | Name of Experiment                                                                                                                                                        | Duration (Hours) | Cognitive Level |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|
| <b>CO1</b>      | <b>Build different diode rectifier circuits</b>                                                                                                                           |                  |                 |
| M1.01           | Plot VI characteristics of<br>1.diode<br>2.Zener diode                                                                                                                    | 3                | Applying        |
| M1.02           | Setup a half wave rectifier with and without filter and plot the input and output waveforms                                                                               | 1.5              | Applying        |
| M1.04           | Setup a centre tapped rectifier with and without filter and plot the input and output waveforms                                                                           | 1.5              | Applying        |
| M1.05           | Setup a bridge rectifier with and without filter and plot the input and output waveforms                                                                                  | 3                | Applying        |
| <b>CO2</b>      | <b>Construct different amplifier and oscillator circuits</b>                                                                                                              |                  |                 |
| M2.01           | Develop an RC coupled CE amplifier<br>(i) Observe the phase difference between input and output waveforms<br>(ii) Plot its frequency response and determine the bandwidth | 3                | Applying        |
| M2.02           | Construct an RC phase shift oscillator and observe the waveforms                                                                                                          | 3                | Applying        |

|            |                                                                                  |   |               |
|------------|----------------------------------------------------------------------------------|---|---------------|
| M2.03      | Construct a transistor astablemultivibrator and plot the waveforms               | 3 | Applying      |
| M2.04      | Construct an astablemultivibrator using 555 Timer IC and verify its output       | 3 | Applying      |
|            | Lab Test – I                                                                     | 3 |               |
| <b>C03</b> | <b>Develop various op - amp circuits</b>                                         |   |               |
| M3.01      | Construct and test Comparator and Zero Crossing Detector circuits                | 3 | Applying      |
| M3.02      | Construct and test Schmitt's trigger                                             | 3 | Applying      |
| M3.03      | Develop an integrator and differentiator circuits and plot their waveforms       | 3 | Applying      |
| <b>C04</b> | <b>Identify various Optoelectronic devices</b>                                   |   |               |
| M4.01      | Demonstrate and study various photodetectors                                     | 3 | Understanding |
| M4.02      | Compare the VI characteristics of different coloured LEDs(Red, Green and Yellow) | 3 | Applying      |
| M4.03      | Demonstrate the VI characteristic of LDR, Photo-Diode and phototransistor        | 3 | Applying      |
|            | Lab Test – II                                                                    | 3 |               |

### Text / Reference

| <b>T/R</b> | <b>Book Title/Author</b>                                                      |
|------------|-------------------------------------------------------------------------------|
| T1         | K.A Navas , Electronics Lab Manual - Vol 1 PHI Learning,ISBN : 9788120351424  |
| T2         | K.A Navas , Electronics Lab Manual - Vol 2,PHI Learning, ISBN : 9789388028080 |

|    |                                                     |
|----|-----------------------------------------------------|
| R1 | V.K.Mehta. Principles of Electronics. S Chand & Co. |
|----|-----------------------------------------------------|

### Online Resources

| Sl.No | Website Link                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Photo diode experiment<br><a href="https://dkpandey.weebly.com/uploads/1/3/5/3/13534845/photo_diode.pdf">https://dkpandey.weebly.com/uploads/1/3/5/3/13534845/photo_diode.pdf</a><br><a href="https://www.youtube.com/watch?v=y3uFMV-B488">https://www.youtube.com/watch?v=y3uFMV-B488</a><br><a href="https://www.egr.msu.edu/classes/ece476/ayresv/Lab08_Fall08.pdf">https://www.egr.msu.edu/classes/ece476/ayresv/Lab08_Fall08.pdf</a><br><a href="https://webstor.srmist.edu.in/web_assets/srm_mainsite/files/downloads/EC1009_electron_devices.pdf">https://webstor.srmist.edu.in/web_assets/srm_mainsite/files/downloads/EC1009_electron_devices.pdf</a> |
| 2     | V-I characteristics of LED<br><a href="https://webstor.srmist.edu.in/web_assets/srm_mainsite/files/downloads/EC1009_electron_devices.pdf">https://webstor.srmist.edu.in/web_assets/srm_mainsite/files/downloads/EC1009_electron_devices.pdf</a>                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3     | <a href="http://www.circuitstoday.com">www.circuitstoday.com</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 4     | <a href="http://wikipedia.org">wikipedia.org</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5     | <a href="#">NPTEL</a> >> <a href="#">Courses</a> >> <a href="#">Electrical Engineering</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 6     | <a href="http://www.electronicsteacher.com">http://www.electronicsteacher.com</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7     | <a href="https://www.allaboutcircuits.com/textbook/experiments">https://www.allaboutcircuits.com/textbook/experiments</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### Student Activity

#### Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation Students should prepare a separate report on open ended experiment of their choice.

Examples.

1. Design a rectifier circuit using Op-amps
2. Using any Simulation software examine the output waveforms of basic circuit
3. Design a 5V dc regulated power supply